QUALITY OF WINE PREDICTION MODEL

a. Problem Statement and understanding (OBJECTIVE)

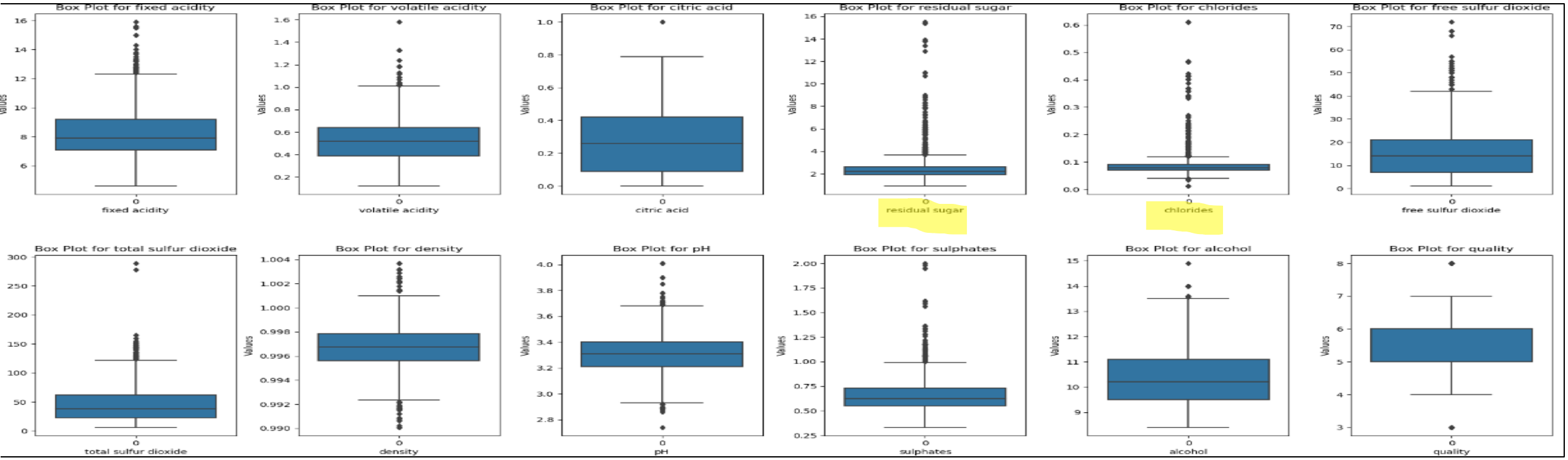
The project that was given was a well-structured data spread sheet of Wine and its different parameters Depending on these parameters the quality of wine is classified from 1-10 , 1 being the low quality and 10 being the best quality
all the parameters used to predict the wine quality are numerical .

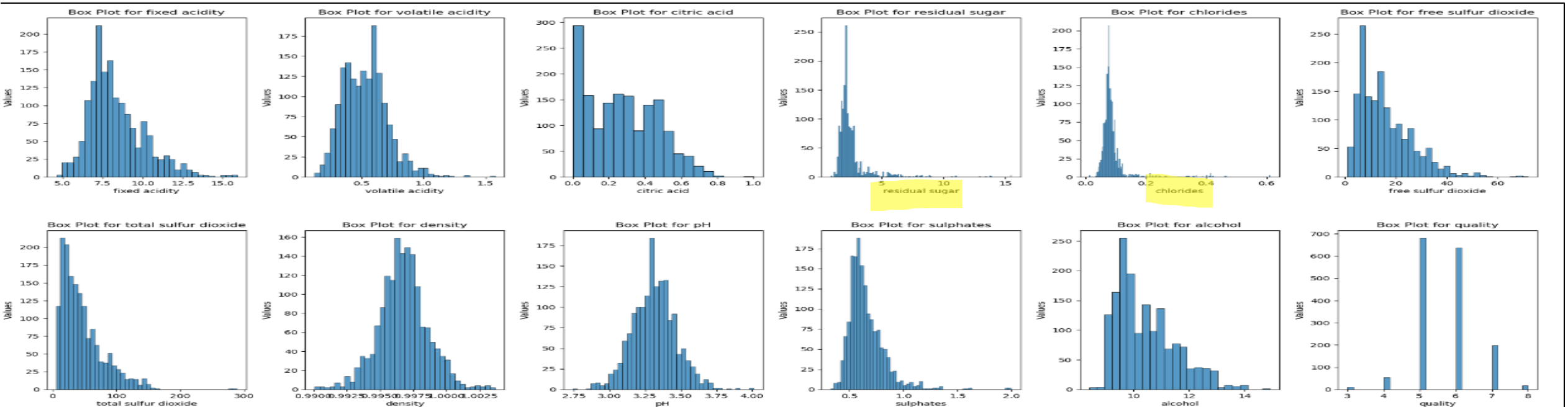
b. EDA Steps Done & Findings (Provide charts, graphs with explanation)

1. First step was to import all the libraries for the data processing and then load the data.
2. Checked for the individual columns and their datatype all infered that all the columns are of numerical data type.
3. Checked for the null values and found none.
4. Plotted a heat map to understand the correlation of the independent variables with the dependent variable and infered that *alcohol* ,*sulphates* ,*citric acid* and *fixed acidity* are more corelated to *quality* than the other independent variable.
5. Checked for the outliers in each independent variables by IQR method also by plotting the boxplot and histogram plot for the same and found that *residual sugar* and *chlorides* has the most number of outliers

```
total number of outliers for total sulfur dioxide :55
percentage of Outliers for total sulfur dioxide based on IQR: 3%
**********************************************************
total number of outliers for density :45
percentage of Outliers for density based on IQR: 3%
**********************************************************
total number of outliers for pH :35
percentage of Outliers for pH based on IQR: 2%
**********************************************************
total number of outliers for sulphates :59
percentage of Outliers for sulphates based on IQR: 4%
**********************************************************
total number of outliers for alcohol :13
percentage of Outliers for alcohol based on IQR: 1%
**********************************************************
```

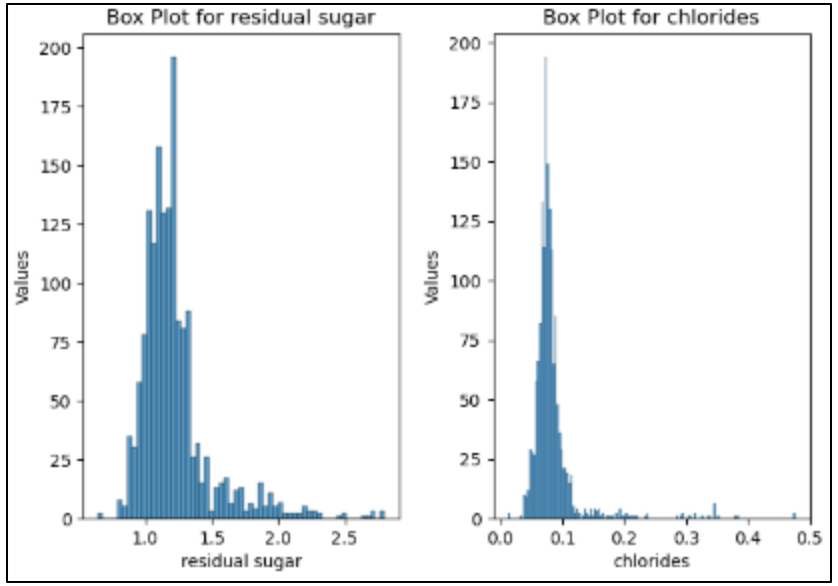
```
total number of outliers for fixed acidity :49
percentage of Outliers for fixed acidity based on IQR: 3%
**********************************************************
total number of outliers for volatile acidity :19
percentage of Outliers for volatile acidity based on IQR: 1%
**********************************************************
total number of outliers for citric acid :1
percentage of Outliers for citric acid based on IQR: 0%
**********************************************************
total number of outliers for residual sugar :155
percentage of Outliers for residual sugar based on IQR: 10%
**********************************************************
total number of outliers for chlorides :112
percentage of Outliers for chlorides based on IQR: 7%
**********************************************************
total number of outliers for free sulfur dioxide :30
percentage of Outliers for free sulfur dioxide based on IQR: 2%
**********************************************************
```


Box Plot for fixed acidity
Box Plot for volatile acidity
Box Plot for citric acid
Box Plot for residual sugar
Box Plot for chlorides
Box Plot for free sulfur dioxide
Box Plot for total sulfur dioxide
Box Plot for density
Box Plot for pH
Box Plot for sulphates
Box Plot for alcohol
Box Plot for quality
Values
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality


Box Plot for fixed acidity
Box Plot for volatile acidity
Box Plot for citric acid
Box Plot for residual sugar
Box Plot for chlorides
Box Plot for free sulfur dioxide
Box Plot for total sulfur dioxide
Box Plot for density
Box Plot for pH
Box Plot for sulphates
Box Plot for alcohol
Box Plot for quality
Values
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality

6. Decided to do the outlier treatment for the 2 columns .Opted treatment is mathematical log treatment and then Plotted the histogram

```
# treating the columns residual sugar and chlorides because they have the most outliers,
# treatment done is log
data['residual sugar'] = np.log(data['residual sugar'] + 1)
data["chlorides"]= np.log(data['chlorides'] + 1)
```



7. Data is then made into a copy so as not to hinder the original data structure

```
# copying the data so as to not hinder the original data
import copy
data_copy=copy.deepcopy(data)
data_copy.head()
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.064711	0.073250	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	1.280934	0.093490	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	1.193922	0.088011	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.064711	0.072321	17.0	60.0	0.9980	3.16	0.58	9.8	6
4	7.4	0.70	0.00	1.064711	0.073250	11.0	34.0	0.9978	3.51	0.56	9.4	5

8. In order to classify the wine quality to good and bad , new list is created for which wine quality of **7 and above is classified as GOOD** and **below 7 as BAD with the header name G_B**

```
G_B= []
for quality in data_copy['quality']:
    if quality>=1 and quality <=7:
        G_B.append(0)

    else:
        G_B.append(1)
```

```
print(G_B[0:10])
```

```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

```
data_copy['G_B']=G_B
data_copy.head(10)
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality	G_B
0	7.4	0.70	0.00	1.064711	0.073250	11.0	34.0	0.9978	3.51	0.56	9.4	5	0
1	7.8	0.88	0.00	1.280934	0.093490	25.0	67.0	0.9968	3.20	0.68	9.8	5	0
2	7.8	0.76	0.04	1.193922	0.088011	15.0	54.0	0.9970	3.26	0.65	9.8	5	0
3	11.2	0.28	0.56	1.064711	0.072321	17.0	60.0	0.9980	3.16	0.58	9.8	6	0
4	7.4	0.70	0.00	1.064711	0.073250	11.0	34.0	0.9978	3.51	0.56	9.4	5	0
5	7.4	0.66	0.00	1.029619	0.072321	13.0	40.0	0.9978	3.51	0.56	9.4	5	0
6	7.9	0.60	0.06	0.955511	0.066724	15.0	59.0	0.9964	3.30	0.46	9.4	5	0
7	7.3	0.65	0.00	0.788457	0.062975	15.0	21.0	0.9946	3.39	0.47	10.0	7	0
8	7.8	0.58	0.02	1.098612	0.070458	9.0	18.0	0.9968	3.36	0.57	9.5	7	0
9	7.5	0.50	0.36	1.960095	0.068593	17.0	102.0	0.9978	3.35	0.80	10.5	5	0

9. plotted a count plot of the dependent variable here G_B and found that the dataset is highly imbalanced so after splitting the dataset to x and y and scaling smoteck is performed on the training dataset for resampling

```
# Balancing the y train
from imblearn.combine import SMOTETomek
```

```
smt = SMOTETomek(random_state = 101, sampling_strategy = 'minority')
X_res, y_res = smt.fit_resample(X_train_scaled, y_train)
```

```
print ("before applying smotetomek=  ", y_train.value_counts())
print ("before applying smotetomek=  ", y_res.value_counts())
```

```
before applying smotetomek=   G_B
0    1267
1      12
Name: count, dtype: int64
before applying smotetomek=   G_B
0    1267
1    1267
Name: count, dtype: int64
```

10. The processed data is then trained and tested using the following models.

- logistic regression (lr)
- knn model (knn)
- decission tree model (dt)
- Random forest (rf)
- xgb boost (xgb)

```
# applying the model

model_lr = LogisticRegression(max_iter = 1000)
model_lr.fit(X_res, y_res)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

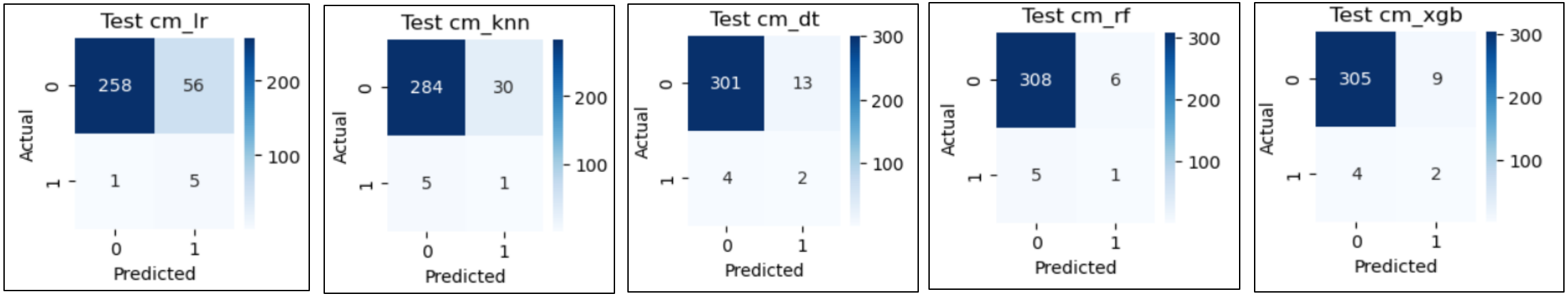
```
#Training Evaluation
y_train_pred_lr = model_lr.predict(X_res)

training_accuracy_lr = accuracy_score(y_res, y_train_pred_lr)
print("Training Accuracy ", training_accuracy_lr)

#Test Evaluation
y_test_pred_lr = model_lr.predict(X_test_scaled)

test_accuracy_lr = accuracy_score(y_test, y_test_pred_lr)
print("Test Accuracy ", test_accuracy_lr)

#Confusion Matrix
cm_lr = confusion_matrix(y_test, y_test_pred_lr)
print (f"confussion matrix={cm_lr} " )
```

11. Evaluation of the models were done on the basis of **Test accuracy, confusion matrix, Recall, Precission and F1_score**

```
tn_lr, fp_lr, fn_lr, tp_lr = confusion_matrix(y_test, y_test_pred_lr).ravel()
print(f"tn={tn_lr},fp= {fp_lr},fn={fn_lr},tp= {tp_lr}")
#Recall
recall_lr = tp_lr/(tp_lr+fn_lr)
print("Recall of Logistic regression model = ", recall_lr)
#precission
precision_lr = tp_lr/(tp_lr+fp_lr)
print("precession of logistic regression model =", precision_lr)
# f1-score
f1_score_lr = (2 * (precision_lr * recall_lr)) / (precision_lr + recall_lr)
print (f"f1 score of logistic regresssion model ={f1_score_lr}")
```

MODEL ANALYSIS RESULT FOR DATASET AFTER *OUTLIER TREATMENT*

CONFUSION MATRIX

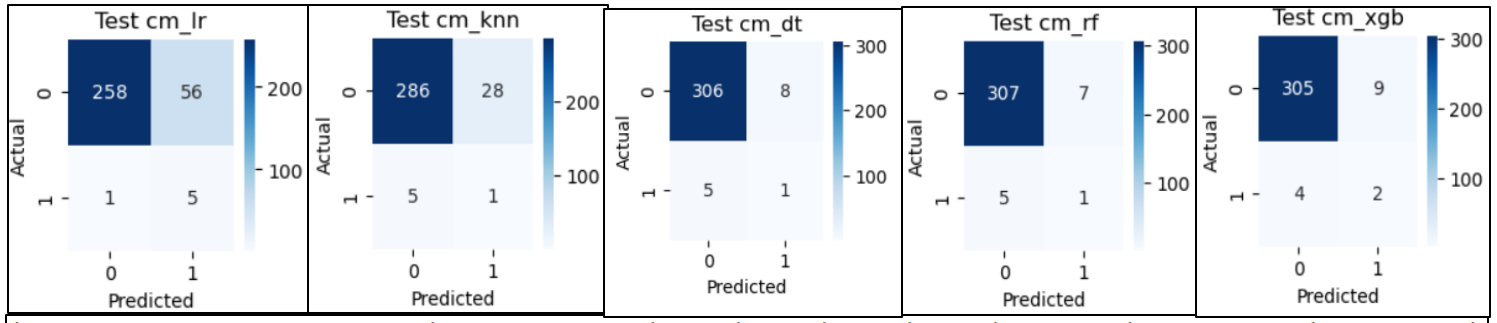


TEST ACCURACY,RECALL, PRECISSION AND F1_SCORE

Algorithm	Test Accuracy	TN	FP	FN	TP	Recall	Precission	F1_score
Logistic Regression Model	0.821875	258	56	1	5	0.833333	0.0819672	0.149254
K-Nearest Neighbors (KNN) Model	0.890625	284	30	5	1	0.166667	0.0322581	0.0540541
Decision Tree Model	0.946875	301	13	4	2	0.333333	0.133333	0.190476
Random Forest Model	0.965625	308	6	5	1	0.166667	0.142857	0.153846
XGBoost Model	0.959375	305	9	4	2	0.333333	0.181818	0.235294

12. the same models where build without outlier treatment but data imbalance was treated and the results were almost same as the above table but slight variation was seen in *Decission Tree Model*

MODEL ANALYSIS RESULT FOR DATASET WITHOUT *OUTLIER TREATMENT*



CONFUSION MATRIX

Algorithm	Test Accuracy	TN	FP	FN	TP	Recall	Precission	F1_score
Logistic Regression Model	0.821875	258	56	1	5	0.833333	0.0819672	0.149254
K-Nearest Neighbors (KNN) Model	0.896875	286	28	5	1	0.166667	0.0344828	0.0571429
Decision Tree Model	0.959375	306	8	5	1	0.166667	0.111111	0.133333
Random Forest Model	0.9625	307	7	5	1	0.166667	0.125	0.142857
XGBoost Model	0.959375	305	9	4	2	0.333333	0.181818	0.235294

TEST ACCURACY,RECALL, PRECISSION AND F1_SCORE

c. INFERENCE:

For the model building of wine quality prediction i infer that the best model solution will be **xgboost model with below accuray and precission after outlier treatment and data imbalance treatment**

XGBoost Model	0.959375	305	9	4	2	0.333333	0.181818	0.235294